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The test consists of **4 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

1.) (essay)

2.) (essay)

a b c d
3.) ☐☐☐☐

4.) ☐☐☐☐

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) When can we say that a set of formulae is unsatisfiable?

Score: 0 / 2.

2.) Define the conjunctive normal form (CNF).

Score: 0 / 2.

3.) Which statements are true for the following three formula?

$$A \Rightarrow q \vee q, \quad B \Rightarrow \neg q, \quad C \Rightarrow q \supset (q \equiv (\neg p \equiv p))$$

(a) $A \Leftrightarrow B$,

-1 $\{A, B, C\}$ is satisfiable,

0 $A, B \models C$,

2 $(A \vee B)$ is valid.

4.) Which statements are true for the following three formula?

$$A \Rightarrow p \equiv q \vee q, \quad B \Rightarrow \neg q, \quad C \Rightarrow \neg(q \supset p) \equiv q$$

(a) $A \Leftrightarrow B$,

2 $\{A, B, C\}$ is satisfiable,

0 $A, B \models C$,

-1 $(A \vee B)$ is valid.